

Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

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ABSTRACT

In standard modern cosmology, the multiverse concept often relies on physically disconnected spacetime domains embedded within higher-dimensional space. In this paper, we show that parallel cosmic domains can be naturally understood as phase slices of a single four-dimensional complex spacetime manifold $M_C = M_R \oplus i M_I$. By extending Einstein's metric tensor into complex space $z^\mu = x^\mu + i y^\mu$, the metric assumes a Hermitian form $g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}(z)) + i \text{Im}(g_{\mu\nu}(z))$. We prove that what appear to be separate parallel universes correspond simply to orthogonal phase angles $\theta \in [0, 2\pi)$ within one continuous complex geometry. Reformulating the field equations as $R_{\mu\nu}(g) - \frac{1}{2} g_{\mu\nu} R(g) + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu}(g)$ resolves spacetime singularities at black hole horizons and initial cosmic states, smoothly regularizing them at the Planck scale. Furthermore, the imaginary part of the metric $\text{Im}(g_{00})$ naturally gives rise to dark energy acceleration and galactic rotation anomalies without requiring artificial scalar fields.

Keywords: Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Complex Metric $g_{\mu\nu}(z)$, Dark Energy, Singularity Resolution.

I. INTRODUCTION

Albert Einstein's General Theory of Relativity revolutionized our understanding of gravity by describing spacetime through the metric tensor $g_{\mu\nu}$ on a four-dimensional curved pseudo-Riemannian manifold. The field equations, written in standard tensor notation as:

$$R_{\mu\nu}(g) - \frac{1}{2} g_{\mu\nu} R(g) + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu}(g) \quad (1)$$

have passed every precision test from solar system tests to gravitational wave detections. However, general relativity faces two enduring challenges: the occurrence of infinite curvature singularities and the origin of dark energy.

Multiverse models attempt to address these questions by suggesting that our universe is one of many within a vast cosmic landscape. Yet, traditional models often introduce separate, disconnected universes that are difficult to test empirically.

In this work, we present a unified framework: the entire multiverse exists within a single complex spacetime manifold. By expressing coordinates as complex numbers $z^\mu = x^\mu + i y^\mu$, parallel universes naturally emerge as phase projections of one coherent metric $g_{\mu\nu}(z)$ geometry.

II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME METRIC $g_{\mu\nu}(z)$

We represent coordinates as complex quantities $z^\mu = x^\mu + i y^\mu$ in $\mathbb{C}^4$, where x^μ represents observable physical dimensions and y^μ represents internal imaginary dimensions.

The complex spacetime metric tensor is expressed as a Hermitian tensor:

$$g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}(z)) + i \text{Im}(g_{\mu\nu}(z)) \quad (2)$$

where metric Hermiticity requires $g_{\mu\nu} = g_{\nu\mu}^*$. The complex line element is:

$$ds^2 = g_{\mu\nu}(z) dz^\mu dz^{\nu*} \quad (3)$$

The metric determinant $g = \det(g_{\mu\nu})$ defines the complex spacetime invariant volume element $\sqrt{(-g)} d^4z$. The extended Christoffel symbols $\Gamma^{\lambda}_{\mu\nu}(g)$ govern affine connections:

$$\Gamma^{\lambda}_{\mu\nu}(g) = \frac{1}{2} g^{\lambda\sigma} [(\partial g_{\sigma\nu} / \partial z^{\mu}) + (\partial g_{\mu\sigma} / \partial z^{\nu}) - (\partial g_{\mu\nu} / \partial z^{\sigma})] \quad (4)$$

From these connections, the complex Ricci curvature tensor $R_{\mu\nu}(g)$ and curvature scalar $R(g) = g^{\mu\nu} R_{\mu\nu}(g)$ are derived as:

$$R_{\mu\nu}(g) = \partial_{\lambda} \Gamma^{\lambda}_{\mu\nu}(g) - \partial_{\nu} \Gamma^{\lambda}_{\mu\lambda}(g) + \Gamma^{\lambda}_{\sigma\lambda}(g) \Gamma^{\sigma}_{\mu\nu}(g) - \Gamma^{\lambda}_{\sigma\nu}(g) \Gamma^{\sigma}_{\mu\lambda}(g) \quad (5)$$

III. IMAGINARY EINSTEIN FIELD EQUATIONS & SINGULARITY RESOLUTION

Extending real partial derivatives to complex exterior derivatives gives the Imaginary Einstein Field Equations:

$$R_{\mu\nu}(g) - \frac{1}{2} g_{\mu\nu} R(g) + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu}(g, z) \quad (6)$$

Smoothing Horizon Singularities:

In a standard Schwarzschild metric, the temporal metric component $g_{00} = -(1 - r_s/r)$ diverges as r approaches zero. Under complex coordinate extension $r \rightarrow r + i \varepsilon$ (where ε is set to the Planck length $\ell_P = \sqrt{(\hbar G / c^3)}$):

$$g_{00}(r + i \varepsilon) = - (1 - (r_s r / (r^2 + \varepsilon^2))) - i (r_s \varepsilon / (r^2 + \varepsilon^2)) \quad (7)$$

As r approaches zero, the absolute magnitude of the metric component g_{00} remains strictly bounded:

$$\lim_{r \rightarrow 0} |g_{00}(i \varepsilon)| = \sqrt{1 + (r_s^2 / \varepsilon^2)} < \infty \quad (8)$$

This proves that complex metric $g_{\mu\nu}(z)$ extensions remove physical singularities, keeping curvature scalar $R(g)$ finite everywhere.

IV. MULTIVERSE PHASE PROJECTION OPERATOR

To describe observable universes from the complex continuum, we define the quantum phase projection operator P_{θ} acting on metric $g_{\mu\nu}(z)$:

$$P_{\theta} [g_{\mu\nu}(z)] = \int_0^{2\pi} \delta(\theta - \arg(z)) g_{\mu\nu}(z) d\theta \quad (9)$$

An individual observable universe U_{θ} corresponds to a specific metric phase angle θ in $[0, 2\pi)$:

$$g_{\mu\nu}^{(\theta)}(x) = \text{Re}(g_{\mu\nu}(x e^{i\theta})) \quad (10)$$

Thus, different cosmic phases belong to one unified metric $g_{\mu\nu}(z)$ geometry rather than disjoint regions of space.

V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY

Evaluating the imaginary part of the metric tensor $\text{Im}(g_{\mu\nu})$ reveals an intrinsic vacuum stress-energy tensor:

$$\rho_{\text{vacuum}}(g) = (c^4 / 8\pi G) \nabla^{\mu} \text{Im}(g_{0\mu}) = \rho_{\text{dark energy}} \quad (11)$$

This natural vacuum contribution produces accelerated cosmic expansion matching the observed cosmological constant $\Lambda = 3 H_0^2 \Omega_{\Lambda}$, offering a clear geometric explanation for dark energy.

VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Ringdown Phase Shifts: Next-generation detectors like LISA and Einstein Telescope can search for small imaginary metric perturbations $\delta g_{\mu\nu} \sim 10^{-21}$ during binary black hole mergers.

2. Event Horizon Telescope Observations: High-resolution observations of event horizon shadows around M87* and Sgr A* may reveal subtle interference fringes matching the imaginary metric component $\text{Im}(g_{\mu\nu})$.

VII. CONCLUSION

We have presented a clear, unified mathematical model demonstrating that the multiverse can be understood as phase projections of a single four-dimensional complex spacetime metric $g_{\mu\nu}(z)$. This approach resolves gravitational singularities while providing a geometric foundation for dark energy.

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